

Airbag-Based Protective System for UAV Medical Payloads

Design and Simulation in LS-DYNA

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by

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Preface

A preface...

*Annamaria Iannuzzo
Delft, October 2026*

Summary

A summary...

Contents

Preface	i
Summary	ii
Nomenclature	vii
1 Introduction	1
1.1 Motivation	1
1.2 Project framing	2
1.2.1 Care centralisation and territorial inequity in the Italian healthcare system	2
1.2.2 Prior research and positioning of the present thesis	2
1.2.3 Stakeholders	3
1.3 Document Outline	3
2 Background and Research-Development Gap	5
2.1 Regional Market Analysis: medical UAV logistics in Piedmont	5
2.2 The <i>payload-first</i> design approach	8
2.3 Airbag impact attenuation: background and design variables	8
2.3.1 Historical perspective	8
2.3.2 Design guidelines	8
2.4 Research Objective	9
3 Methodology - Airbag Design Workflow	10
3.1 Problem statement	10
3.2 Proposed approach	12
3.2.1 (Outer) design loop formalisation	12
3.2.2 Proposed approach	12
4 Results - Through the Workflow	13
4.1 Mission and failure scenario analysis: calculating V_x and V_y	13
4.2 Studying the payload: calculating allowable impact acceleration	13
4.2.1 Material characterisation	13
4.2.2 Sled test	13
4.3 Design variables optimisation: the code!!!	13
4.4 Summary of results	14
5 Conclusions	15
5.1 General summary	15
5.2 Recommendations for future work	15
6 Project Management	16
6.1 Strategy and planning	16
6.2 Challenges	16
6.2.1 challenge 1: aligning interests for thesis scope	16
6.2.2 challenge 2: Computational resources NO MORE	16
6.2.3 challenge 3: IT infrastructure bottleneck	16
6.2.4 challenge 4: going against the judgement of supervisors	16
6.3 Lessons learned	16
References	17
A Extended Market Analysis	21
A.1 Value-chain actors	21

A.2 Trials and planned initiatives	24
B FEM Iterations for glass phial model	26
C Task Division Example	27
Acknowledgements - <i>Ringraziamenti</i>	28

List of Figures

List of Tables

2.1	Actors involved in drone delivery of medical goods in Piedmont, grouped by value-chain stage.	6
A.1	Actors involved in drone delivery of medical goods in Italy	21
A.2	Drone delivery of medical goods in Italy: trials and planned initiatives	24
C.1	Distribution of the workload	27

Nomenclature

If a nomenclature is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Abbreviations

Abbreviation	Definition
ISA	International Standard Atmosphere
...	

Symbols

Symbol	Definition	Unit
V	Velocity	[m/s]
...		
ρ	Density	[kg/m ³]
...		

Introduction

1.1. Motivation

The present MSc Thesis in Aerospace Structures and Materials falls under the category of *Engineering Design*, in alternative to the *Scientific Research*-type projects. In *design* projects, it is required to attempt the process of innovation not through the discovery and exploration of a research niche, but rather by applying pre-existing scientific knowledge to enable the concrete realisation of a technical design. In the case of the present thesis, the overall objective consists in proposing a *design workflow* and validating it through testing and simulation. If adopted in the preliminary design phase of a drone-based delivery mission, this workflow yields an optimised airbag-type device tailored to a specific failure-and-drop application, in line with the requirements set by the project commissioners. Therefore, the successful completion of this thesis is achieved by:

1. Proposing an application-specific design workflow that fits within the broader context of the ongoing research effort;
2. Testing the workflow, both experimentally and through simulation, to realistically inform the device's design.

The design workflow (informally called *inner (design) loop* throughout this thesis) is intended as a first attempt to bridge the gap between the long-term, broadly-defined objective of constructing a large-scale drone network (also called *outer (design) loop*) over the region of Piedmont, Italy, and the previously completed research work, consisting in a series of logistic, legislative, and generally high-level evaluations of possible **drone-based missions** with the objective to **deliver medical supplies in emergency scenarios**. Such work was conducted *Politecnico di Torino*, in collaboration with the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex and automotive consulting company *FEV Italia S.r.l.*, three of the main industrial and research stakeholders working towards helping hospitals in the transition from road to air transport to enhance the logistics of the *Sistema Sanitario Nazionale (SSN)*, the Italian National Healthcare System) both at an intra- and inter-hospital level.

This MSc Thesis Project is, therefore, a multidisciplinary project where the knowledge and techniques of Structural Engineering are put to the service of the medical field. The airbag-like device resulting from the design process should be able to protect the free-falling medical payload in the event of a drone failure and its subsequent detachment from the latter. The device aims to bring the impact accelerations within the allowable limits of the payload container and of the medical payload being transported, in compliance with the safety regulations regarding dispersion of chemical substances into the environment. However, contrary to most aeronautical projects which do not distinguish between the integrity of the aerial vehicle and the safety of its passengers, this project adopts a "payload-first" approach where the protection of the highly valuable medical cargo is prioritised over the integrity of the drone carrying it. Distinguishing between the safety aspects of the unmanned aerial vehicle (UAV) and its payload implies a certain facilitation in the certification process of the UAV, as well as for the modular implementation of the safety device on a variety of UAVs serving in different, specific delivery

missions.

1.2. Project framing

1.2.1. Care centralisation and territorial inequity in the Italian healthcare system

The idea of transforming healthcare delivery networks from road- to air-transport-based systems stems from the consensus within the Italian medical field, consolidated during the public health emergency of the COVID-19 pandemic (2020–2022), that the current ambulance-based architecture of the SSN's logistic infrastructure (described in detail¹ in [1]) cannot accommodate the healthcare system's needs and transitions. The two main - and competing - factors are:

1. **Care centralisation:** [2] describes care centralisation as the ongoing process of concentrating medical expertise and services in key geographical areas, e.g. provincial and/or regional capitals, to meet the demands of high-volume care. The downside of care centralisation is that at a regional level, it induces "distance-related inequity, particularly for rural and peripheral populations", especially heightened in emergency scenarios.
2. **Unhomogeneous distribution of emergency services :** [3] affirms that, in contrast with the centralisation maneuver described above, emergency resources are still unhomogeneously distributed over the regional territories, with services overloaded in densely populated areas and underused in low-density ones, resulting in inefficient overall utilisation.

In conclusion, [4] builds on these two points by affirming the importance of establishing a generally more efficient strategy for healthcare management at all levels, and incentivises the development of technologies suitable for upgrading, among the several aspects of the SSN, the logistics of the Italian healthcare supply chain.

1.2.2. Prior research and positioning of the present thesis

In response to the issues highlighted in Section 1.2.1, recent years have seen an increasing number of technology development initiatives where industry players in the transport sector and research institutions work together to push for the modernisation of transportation means within the Italian SSN. Several of these industry-research partnerships have ventured into the globally booming market of unmanned aerial vehicles, where remotely-piloted or fully autonomous drones are being used ever more often to carry medical payloads in a plethora of scenarios: intra- vs. inter-hospital delivery [5], ordinary logistics vs. emergency missions and humanitarian applications[6], experimental/direct flight between strategic poles [7] vs. a fully-operational, inter-hospital, hub-based UAV network covering national territory (the *Zipline* case study[8]).

Following the global trend to answer a local need, in the specific case of the Piedmont region in north-west Italy, where at the time of writing there exists no city- or region-wide UAV network deployed for the delivery of medical goods, many pilot programmes (further analysed in Section 2.1) are emerging, among which a partnership between the Pharmacy department of the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex (head-quartered at Molinette Hospital in Torino), a research group within the Department of Mechanical and Aerospace Engineering at *Politecnico di Torino*, and the Numerical Analysis Team of *FEV Italia S.r.l.*, the Italian branch of the German-founded multinational company *FEV*, a global leader in research and development for the automotive industry.

The six-year-long partnership between these three institutions is the backbone of the present MSc thesis project, the most recent of a chain of research projects aiming to pursue the long-term vision of flanking, if not completely replacing, the current road-based transportation means with aerial alternatives within hospitals in Piedmont and, specifically, in the city of Torino (the region's capital), in favour of more rapid, more efficient, and safer transportation of medical supplies. Specifically, this research effort has broadly tackled the key aspects required to establish a complete and regulation-compliant UAV medical delivery ecosystem in the Piedmontese territory, resulting in five MSc theses and one conference paper presenting the following results:

¹"The type of vehicles used varies from region to region. In Piedmont it mainly consists of ambulances used only for transport plus more advanced vehicles equipped for resuscitation, replaced in some sites by mSa on a car with a doctor, nurse and rescuer driver, organized on the basis of specific needs of the territory, different in the four macro-areas of the emergency network (rEmE) in which the region was divided." [1]

- **Cavallo (2020)** [9]: Preliminary assessment of UAV platforms suitable under Italian and European regulations. Development of risk maps and definition of optimal trajectories between selected hospitals.
- **Bertelli (2021)** [10]: Detailed selection of UAV based on mission specifications derived from the medical goods to be transported and ENAC regulatory constraints. Application of the SORA (Specific Operations Risk Assessment) methodology to three feasible mission routes, yielding the safety requirements that operators must satisfy.
- **Sivieri (2021)** [11]: Identification and characterisation of thermal and vibrational loads encountered during typical UAV medical missions. Experimental testing on milk-derived sera samples to determine their resistance to UAV-induced vibrational stresses. First hypothesis of medical payload container geometry and materials, implemented in the experimental campaign.
- **Fiorentino (2021)** [12]: Development of a path-planning strategy for life-saving drug distribution in populated environments. Generation of risk maps and implementation of an RRT-based algorithm to minimise global risk and flight time.
- **Angelini (2023)** [13] and **Angelini (2024)** [14]: Comparative analysis of the safety and economic efficiency of emergency medical deliveries performed by car, drone, and VTOL aircraft. Identification of optimal pharmaceutical delivery hub locations for each transport mode to maximise logistical performance.

The present thesis was carried out in collaboration with the three institutions mentioned above to advance the research effort, bridging the gap between their long-term objective and the results achieved so far. This required preliminary work aimed at formalising the project requirements at both a broad, high level and at a specific, first-step level. Having taken a legislative, logistic, market- and mission-evaluation approach until now, it was possible to delineate a scenario in which the approval of ENAC (the Italian Civil Aviation Authority - *Ente Nazionale per l'Aviazione Civile*) could potentially be obtained to enable this type of operation. However, as certifications are unavoidably specific to mission type, trajectory and vehicle, it appeared necessary to implement a more concrete approach rooted in *design*, whilst also adopting a "payload-first" approach that diverted the project towards a device for the *safety of the medical payload* rather, than that of the aerial vehicle. The deliberate prioritisation of this aspect is justified in Sections 2.1 and 2.2, where it is illustrated on the basis of a regional market analysis and the exploration of an ongoing paradigm shift in the design process of biomedical devices.

1.2.3. Stakeholders

Overall, the stakeholders participating in the present project had the following roles:

- **TU Delft** (Prof. Bianca Giovanardi): Primary academic supervisor overseeing the scientific quality and methodological development of the research.
- **Politecnico di Torino** (Prof. Enrico Cestino, Davide Angelini): Institutional leads and co-supervisors for the broader UAV medical-delivery research project, providing access to the experimental platforms used.
- **A.O.U. Città della Salute e della Scienza di Torino / Molinette Hospital**: Provider of medical and operational data related to pharmaceutical transport requirements and mission scenarios.
- **FEV Italia S.r.l.**: Industrial partner offering technical support for advanced modelling and simulation activities, including SolidWorks-based design and LS-DYNA impact analyses.

The interaction between stakeholders, as well as the management and communication aspect of the project, is analysed in Chapter 6 - "Project Management".

1.3. Document Outline

DISCORSIVO. Esempio:

The thesis is structured as follows, first, chapter 2 presents a literature review to outline the several possibilities to predict the buckling load of cylindrical shells and which techniques are available to reduce the sensitivity to imperfections. The chapter concludes by identifying the research gaps related to the buckling of cylindrical shells. From this conclusion, the research objective and scope are defined

in chapter 3. The goal is to include imperfections within the displacement-based Koiter methodology and the necessary steps to achieve this are explained in chapter 4 to 7, as well as how effective this novel modelling technique is. Finally, the concluding remarks can be found in chapter 8 and future research is presented in chapter 9.

2

Background and Research-Development Gap

In technology development, it is important to gain a clear idea of two aspects:

1. **Research aspect:** if the technology that is being developed requires the exploration of an uncharted research niche, as opposed to utilising pre-existing and consolidated knowledge for the creation of a new product;
2. **Development aspect:** whether the proposed technology targets a well-defined market with unmet customer demand and no established supply.

This chapter explores both these aspects in Sections 2.2-2.3 and 2.1, respectively. The aim is to gather sufficient background information, from market allocation of the proposed technology to concrete understanding of airbag design criteria, to depict the choices proposed in Chapter 3 regarding operative *methodology* as an answer to the Research Questions proposed at the end of this chapter.

2.1. Regional Market Analysis: medical UAV logistics in Piedmont

The Italian National Healthcare System (SSN) is structured hierarchically: the guidelines and regulations issued at a *national* level must be implemented locally, at a *regional* level, by each region's *Sistema Sanitario Regionale* (SSR, the Regional Healthcare Service). Vice versa, aviation rules follow European-level enforcement mechanisms: Following the 2019 harmonization of European drone operations across the "open" and "specific" categories [15], operators seeking authorization for high-risk flights are required to conduct a risk assessment using the Specific Operations Risk Assessment (SORA) methodology developed by JARUS¹, making a thorough evaluation of potential failure scenarios while proving compliance with the demanding safety standards imposed by aviation authorities.

The discrepancy between *modi operandi* of the aviation and healthcare authorities has shaped the way drone-network operations have mostly been developing in the last decades in Italy: several emerging, albeit scattered, research groups, either cooperating or competing with industrial counterparts, have achieved geographically limited trial deployments while being unable to scale the operation beyond testing.

A complete summary of such operations, limited to the means of investigation available for the present thesis project, can be found in Table A.2 in Appendix A.2. The table shows a clear trend: although potentially booming as several healthcare facilities are exploring the possibility of adopting drone transport and operators are proposing new drone models for them, most activities stop after the initial demonstrations, with the only exception being the region of Abruzzo where the operational phase of the drone drug-delivery service between a strategic hospital and village pharmacies was announced in July 2025 [16], [17]. In general, a pattern of fragmentation, short-lived pilots, and few transitions

¹Joint Authorities for Rulemaking on Unmanned Systems

to a potentially scalable service is observed in the construction of the overall *value chain*, with many independent institutions covering equivalent roles, as shown in Table A.1 in Appendix A.1.

Table 2.1: Actors involved in drone delivery of medical goods in Piedmont, grouped by value-chain stage.

Category	Institution	Scope
Drone manufacturers	SkyEU	Autonomous high-speed eVTOL drones for first aid and critical logistics [18]
Technology partners	Mavtech	INDOOR technical partner; aerial surveillance and operational-support products [19]
	FEV Italia	Industrial partner of the DIMEAS feasibility studies; numerical impact analysis
Payload / container makers	ABzero (Pisa)	Smart Capsule used for the INDOOR flight: sensorised container for blood, organs and drugs [20], [21]
Operators	ABzero (Pisa)	Supplied the drone and handled the ENAC authorisation for INDOOR [22], [23]
Airspace and infrastructure	Torino Caselle and Cuneo Levaldigi airports	Proposed hub and support base for hydrogen drone logistics [24]
Research	PIC4SeR, Politecnico di Torino	Autonomous flight technologies for transporting biological material (INDOOR) [25]
	DIMEAS, Politecnico di Torino (Prof. E. Cestino)	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost-safety comparison of car, drone and eVTOL, hub siting [9], [10], [11], [12], [13], [14]
	Università di Torino	INDOOR project partner [19]
End users	AOU Città della Salute e della Scienza	INDOOR test flight between hospital sites; proposed hydrogen logistics plan; pharmacy department partner of DIMEAS feasibility studies [21], [24]
	Regional Transplant Centre Piemonte–Valle d’Aosta	INDOOR partner for organ and biological transport [19]
Promoters and funders	Fondazione DOT	Promoter of the INDOOR project [19]
	Centro Nazionale Trapianti	Funds the INDOOR research grant at PIC4SeR [22]
	City of Turin	INDOOR partner; funded the LastMed call [19], [26]
Regulators and authorities	ENAC	National civil aviation authority; INDOOR partner and flight authorisation [19]
	Regione Piemonte	Regional health authority; drone trials subject to benefit–risk analysis [27]
Alternatives	Tecnocad / Tecnomill / Bylogix	LastMed autonomous ground rover for last-mile drug delivery [28]

Adopting a region-based, rather than a nation-based, approach for market analysis is, therefore, necessary. Table 2.1 highlights stakeholders involved in Piedmont-based operations and their roles in the value chain, illustrating a fragmented scenario: Project INDOOR, launched by Fondazione DOT in 2021, culminated in a 500 m-long test flight by an autonomous UAV between two Città della Salute hospital sites, carrying medical goods in a dedicated capsule, on 1st October 2024. In this project, several other stakeholders were involved (in fact, most of the ones present in Table 2.1), among which

ABZero (a spin-off from *Scuola Normale Superiore di Pisa* responsible for capsule design and handling the ENAC authorisation process for the test flight), parallel research group PIC4SeR from the Department of Electronics and Telecommunications at *Politecnico di Torino*, and MAVTech S.r.l. (UAV systems designer company based in Torino). Independently of the INDOOR project of 2021-2024, the startup SkyEU was founded in 2025 promising to develop eVTOL drone systems for critical logistics within, but not limited to, the healthcare sector.

In parallel, and independently of INDOOR, the research group led by Prof. Enrico Cestino at the Department of Mechanical and Aerospace Engineering (DIMEAS) of *Politecnico di Torino*, together with A.O.U. *Città della Salute e della Scienza* and FEV Italia, has pursued since 2020 a series of feasibility studies on the Molinette–San Giovanni Bosco route (Section 1.2.2). These covered regulatory and mission evaluation, risk mapping and path planning, payload environmental loads, and cost–safety comparisons between road, drone and eVTOL transport. No flight campaign has been carried out so far. This makes the group the only Piedmontese actor whose work so far has been entirely at the planning and assessment level.

On the other hand, the regional administration’s general stance with respect to this type of operation highlights the fundamental requirement of thorough benefit-risk analyses to guarantee both the safety of the Piedmontese citizens and the preservation of the medical cargo, possibly needing to limit transportation to exceptional emergencies in mountain areas [27]. From an engineering perspective, these criticalities can be translated in the following barriers for the development of a full-scale, UAV-based delivery network:

1. **Authorisation of flights:** Operations beyond the open category are approved by compiling the SORA. Among the different classes of mitigations, *ground risk* mitigations (one of the first steps required in the SORA) reduce the consequences of the UAV’s impact on the ground after a loss of control, for example by introducing a parachute. It is key for technical stakeholders to prove the reliability of their implemented ground risk mitigations to protect civilians in case of drone failure.
2. **Payload integrity and containment:** In accordance with the regional administration’s concerns, current certification standard UN3373 for ground handling of Biological Substances, Category B, establishes that packages must pass a drop test at 1.2m with no leakage from the primary container **CITE STANDARD HERE**. However, in the event of a free-fall from ordinary mission altitudes, this standard loses its significance and the potential for biological material dispersion into the environment becomes highly relevant.
3. **Unsustainability of the current business models** While costs of road and helicopter transport of biomedical goods such as organs, drugs and other critical supplies are expected to fall significantly with drones, most trials are funded by EU projects and one-time grants, with the consequence of making the project unsustainable and non-scalable.

On the basis of this analysis, several research and development gaps can be identified: SkyEU is endeavouring to enact an economically sustainable business model for large-scale UAV networks, but has a small public record; ABZero has patented capsules equipped with electronics that reliably monitor in-flight vibrations and guarantee thermal control of the medical payload, but these are only certified to the UN3373 standard, whereas in case of in-flight failure a much more conservative approach is needed; regional authorities need reassurance on the safety aspect for civilians, as a 500 m test flight on a controlled route shows that the obstacle is not drone-making, but guaranteeing safety in everyday scenarios rather than in single monitored missions.

In this scenario, the institutions (*DIMEAS – Politecnico di Torino*, *FEV Italia*, *AOU Città della Salute*) that commissioned the present thesis project aim to tackle, among the three barriers, barrier 2, concerning payload integrity and containment. The proposed solution is an airbag-type device that protects the payload and prevents its dispersion in the event of an impact, especially in view of transporting extremely valuable medicinals and/or biological materials, e.g. human organs for urgent transplant operations. This device is conceived as an add-on to existing capsules such as the one by ABZero, mounted on drones such as SkyEU’s, and is intended to help close the certification gap highlighted above, while acknowledging that it does not address the SORA ground risk or the sustainability of the business model.

2.2. The payload-first design approach

To justify this paragraph: Paradigm shift for design, For all aeronautical projects, Which might be the common denominator for their failure in implementing. It might be a suggestion for future aeronautical projects, in general. That, to guarantee the success of the projects, safety really needs to come first, just like in biomedical projects.

Chhetri - an updated REVIEW of medical drones in 2026 * overcome geographical barriers and provide critical care in remote or disaster-stricken areas. * critical role of payload container design, including fixed, cable-suspended, and temperature controlled systems, and analyze its effect on both the medical supplies and the overall flight dynamics of the aircraft. * These supplies may include vaccines, medications, blood units, diagnostic samples, and organs for transplantation.

regarding the payload container: "It must also meet safety standards, ensuring dust tight sealing and protection against strong water jets, thus shielding the container and its contents from mechanical impacts and external forces. Aerodynamic efficiency is another vital consideration as it lessens the power necessary for flight. Additionally, the choice of materials is crucial to the container's performance, with lightweight yet sturdy materials selected to preserve structural integrity while preventing unnecessary energy use. Finally, the transport container must be operationally compatible, providing flexibility in deployment across various aerial transport platforms." does not mention protection of what is inside.
— CHECK REFERENCE 25

* drone is designed first, then "put a parachute on it" (the safety requirement by Alulema, which Mass feedback loop: a heavier aircraft needs a bigger parachute, which makes it heavier. This is why the requirement costs 36 percentage points of feasibility (Table 1: 84* focus on vibration isolation and temperature isolation, keeping the cold chain * general impression: that the impact on aerodynamics and flight mechanics is included before even thinking about the integrity of the payload in case of impact.

PUT A BOX HERE — the idea of the airbag which tackles limitation 2 presented earlier, combined with the idea of preserving the payload.

WHAT IF WE THOUGHT ABOUT REVERSING THE DESIGN PROCESS? Design the drone AFTER having hypothesised the safety device that is included in the payload container in the first place? QUOTES: "Success is measured at the patient, not at the device." Success is measured at the vial, not at the airframe. A drone that survives while the payload arrives cracked has failed the mission. The same goes for a device that works but doesn't serve the patient.

Rodriguez Drawing a parallel to the medical tech design approach, which is often something that medical tech designers need to be reminded of when designing the coolest tech for patients. "Ashift in the traditional technocentric view of medical device design to a human-centered one is needed to bridge existing translational gaps and improve healthcare. called "translation gap" or "implementation gap"

in this case, the gap is between the technological potential (the drones) and 2 ends, i.e. the normative safety constraints where a parachute is not enough, and, most importantly, protecting the payload which is arguably more valuable than the drone itself.

2.3. Airbag impact attenuation: background and design variables

2.3.1. Historical perspective

* Automotive Engineering

* Space Engineering - implemented in impact applications, similar to the one requested for this case, since the 1960's for xxx mission. - analytical, hard-codable solutions are 1Dof and rare, FEM is immediately chosen to include 2- and 3d effects of impact.

however, both fields have the guidelines in common, which are explained in the following paragraph

2.3.2. Design guidelines

* Avendo un panorama più chiaro su quella che è l'obiettivo a Torino, si è scelto di concentrarsi sulla mitigazione del ground risk whatever. La scelta ricade su una tipologia di dispositivo di sicurezza atto

a mitigare il rischio di impatto, danneggiamento delle fiale, dispersione nell'ambiente, etc. un Airbag

Perché è stato scelto l'airbag? paracadute: * pesanti (spessi) * non reggono impatto * paracadute è soltanto sul drone - medicinali ad ogni modo persi * sgancio + paracadute meglio di sgancio + airbag — obiettivo, protezione medicinali, anche se il drone viene protetto tanto meglio * vantaggi: può essere usato in ospedale (basse quote) * in acqua ha più senso impermeabilizzare l'airbag e non il contenitore, con il vantaggio che può stare effettivamente a galla

Stabilito che l'airbag è la tecnologia su cui ci vogliamo concentrare, allora le cose da prendere in considerazione sono: lista delle cose del documento in cui ho preso appunti.

Variabili da prendere in considerazione quando c'è da progettare un airbag

Phase 1 variables for peak acceleration 1. Shape 2. Size 3. Fabric 4. Vent size 5. Vent burst pressure 6. Inflation pressure

Phase 2 variables for deployment time 1. Housing/container size and geometry 2. Air injection inlet pressure 3. Air injection mass flow rate 4. Seams locations 5. Folding pattern 6. Number of folds

2.4. Research Objective

Having presented the context in which the research project is allocated and having gathered the necessary design information, it is possible to address the requirements set by the project commissioners:

- Identify and characterise the mechanical allowables of typical medical payloads based on general packaging requirements for UAV-transported medical goods.
- Develop an informed conceptual design of an airbag system capable of deploying during emergency release conditions to absorb impact energy and preserve payload integrity.

These requirements can be summarised in the following Research Objective and Research Questions:

Research Objective

Develop and validate a design workflow to model and simulate an airbag-based protective system for UAV medical payloads using LS-DYNA, with the primary objective to ensure the structural integrity of pharmaceutical materials and their containment during drone failure and emergency payload release, thereby preventing environmental dispersion and supporting certification requirements.

Research Question(s)

How can a systematic multi-phase simulation workflow be developed to design an airbag-based protective system for medical payloads during UAV emergency-release events?

1. What are the allowable mechanical limits (acceleration, stress, etc.) for medical products during emergency release, and how should these define the design envelope?
2. How can the inflated airbag configuration be optimised to satisfy impact-absorption requirements while balancing competing design requirements?
3. How can the optimised inflated geometry be transformed into a compact and functionally deployable system?

3

Methodology - Airbag Design Workflow

3.1. Problem statement

What I have been given, hypothesis of payload (Akineton Fiale, but most importantly the fact that they are simply exemplary phials) and payload bay (designed from former research) * I focus on the inner design loop with safety, dictated by SORA

* Dato che fino ad ora il problema è stato normativo, si è partiti da un approccio di calcolo della probabilità di failure. Ci sono state 5 tesi e svariati lavori scientifici con l'obiettivo generalizzato di quantificare il rischio di un'operazione di droni che sorvolano la città, per poi passare questo testimone a futuri ricercatori con l'obiettivo di MINIMIZZARLO (anche se alcune indicazioni sono state già effettivamente rilasciate)

1. Individuazione SCENARI DI MISSIONE - tesi BERTELLI 2021 * Specific Operations Risk Assessment (SORA) - adottata da EASA, consiste in: * Definizione dei Concept of Operations - dove abbiamo la DESCRIZIONE DELLA MISSIONE, che non si può cambiare * Determinazione della Ground Risk Class (GRC) - ha un certo valore 6, pg. 39, per diminuirlo bisogna implementare metodi di MITIGAZIONE (3 tipologie), io agisco nell'ambito della mitigazione M2 (pg. 42, mitigare effetti dell'impatto al suolo). PG. 46!!!! M2, progettazione tecnica. Nuova valutazione del GRC pari a 3 (pg. 54) * Valutazione dell'Air Risk Class (ARC) - collisione con altri mezzi nello spazio aereo, fuori dal mio scopo di tesi

* Primatista (ref. 8 e 9) - ispirazione anche per Davide * l' Ospedale Molinette per l'AOU Città della Salute e delle Scienze e l'Ospedale San Giovanni Bosco per l'ASL Città di Torino - poli ospedalieri più fondamentali per la missione (posizionati ai lati opposti della città) (pg. 23) * requisito del drone: deve essere già equipaggiato di paracadute (pg. 23)- failure del paracadute è lo scenario in cui mi colloco io. " Il paracadute può decelerare il carico utile fino ad una velocità di 3 m/s, al fine di fornire un atterraggio sicuro." pg.19 —> 3m/s causa comunque certe accelerazioni di impatto non indifferenti, che non garantiscono l'integrità dei medicinali. —> simula anche a 3m/s v verticale????????? CASO ACQUA E CASO TERRA???

* drone scelto: Matrice 300 RTK della casa produttrice DJI (che comunque deve essere equipaggiato di paracadute) - pg. 25 * capacità massima di carico che equivale a 2,7 kg - LIMITE MASSIMO ENTRO CUI IL MIO PROGETTO (PARACADUTE, AIRBAG, CONTENITORE) DEVE STARE. * Resistenza al vento: 15m/s in crociera, <https://www.dji-store.it/prodotto/dji-matrice-300-rtk/?srsltid=AfmBOorTcxO1yqTrZgryaDUUjwIWj> <https://www.eliteconsulting.it/store/dji-matrice-300-rtk/>

* la scelta del drone non è univoca e "Ed anzi, al momento della realizzazione effettiva del progetto, sarà prevista la progettazione di un UAV "ad hoc", che verrà progettato e realizzato col solo ed unico fine di condurre questa specifica missione" - pg. 25

* pg 31 - "il drone mantenga durante tutta la tratta prestabilita una velocità di crociera di 17 m/s e che si

nuova ad una velocità pari a 5 m/s per effettuare la fase di decollo verticale fino alla quota di 75 m" * fig 4, pg. 32 caso 1, fig. 5 pg. 33 caso 2 (leggermente più pericoloso, ma comunque) - tratta sul PO che collega i due ospedali, 15 minuti di tragitto, 10km - paragonabile al tempo percorso su gomma (fattore traffico, anche se saremmo in ambulanza a sirene spiegate)

* COLLEGAMENTO CON TESI DI DAVIDE - i suoi indici di rischio dicono che non è fattibile una missione con drone sul territorio cittadino —> quindi andiamo sul PO

* possibili velocità di crociera: $v = 17\text{m/s}$ realistica, 23m/s v massima realizzabile dal drone (pg 35)

* Altezza operativa massima prevista: 75 m. * MTOM ("Maximum Take Off Mass") del drone: 9 kg. * Dimensione massima caratteristica (d) dell'aeromobile: 0.81 m. * Velocità (v): 17 m/s.

* tabella 8 pag. 67

* terminal velocity calculator that I can use too: <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/termvel/>

* rischi legati all'ottimizzazione dell'ultima tratta su terra (vari scenari: linea retta, un parco, etc.) — da considerare?

TESI CAVALLO 2020 * Individuazione tratta Molinette - San Giovanni Bosco (pg. 15) * Distinction between two types of transportation: Hospital-to-Hospital, Hospital-to-Home (pg. 20)—> my case is the first one. * Wind information: the city of * idea originale del corridoio acquatico soddisfa i requisiti normativi restringenti dell'enac (pag. 37)

Caratteristiche del drone: * Cruise speed in m/s; —> 70 • Maximum flight time in minutes; —> half of drone's autonomy • Mass in kg; • Radius in m; • Frontal Area in m²; • Glide speed in m/s; —> calculated differently for each drone type (pg 43) * WIND —> average of 2m/s throughout the year, 23m/s max.

* two types of descent: ballistic descent and uncontrolled glide are already mentioned here * first to suggest the river corridor solution —> brings to Bertelli

2. CONTENITORE DA PORTARE - trasporto dei medicinali, tesi SIVIERI 2021 * pg 19 sivieri: "Il tronco centrale è a sezione cilindrica costituito da 4 strati. Partendo dallo strato interno e procedendo a quello più esterno: 1. Strato interno di materiale PETG di spessore 2 mm; 2. Strato di materiale isolante di natura e spessore incognito; 3. Strato esterno di materiale PETG di spessore 2 mm; 4. Strato di vernice termo riflettente. • Le estremità sono costituite cupole emisferiche di costruzione equivalente a quella del tronco. * progettato per soddisfare requisiti termici e vibrazionali di una missione tipica * CARATTERISTICHE (pg. 27 - 31 SIVIERI) * messa in tavola fig. 2.12 * spessore di 2 mm sulla parete esterna ed interna, con un riempimento interno dell' 1 % con pattern rettangolare, come in 2.15 * la cavità interna è stata riempita di schiuma poliuretanica (scelta di iniettabilità efficace) * superficie esterna è stata scartavetrata e sono state applicate tre strati di vernice argentata (2.16) per migliorare la riflettanza del materiale

* MEDICINALI DA TRASPORTARE - PARENTESI SULLE TIPOLOGIE DI CONTENITORI E SCATOLE DI FARMACI QUI

3. INDICI DI MERITO BASATI SUL RISCHIO - TESI ANGELINI 2023: * Metodo per determinare la traiettoria di rischio minore: determinare la traiettoria di minimo rischio tra due punti si usa l'algoritmo di Dijkstra e l'algoritmo A* (del quale Dijkstra è un caso particolare). pg 17 * Mappe di input: densità della popolazione (pg. 35), mappa dei "ripatri" (ref. 17 nella tesi, pg. 40) cioè degli edifici che rappresentano un fattore di sicurezza per il sorvolo della città (stima di vari fattori di protezione ispirati a diversi lavori scientifici)

* modi di caduta più frequenti: ballistic descent (forza di gravità e resistenza aerodinamica, eq. 3.3 - risolta numericamente per trovare velocità e punto di impatto al suolo, tabella 3.3) and uncontrolled glide (ref 17 in tesi) —> ballistic descent: velocità x iniziale VTOL = 140m/s, velocità del vento media 25.2 m/s, —> uncontrolled glide: * tempo di impatto al suolo eq. 3.11 * distanza percorsa eq. 3.10 * mappe di rischio generate con metodi probabilistici * CALCOLO DEL PERCORSO OTTIMO focus drone: —> traiettoria di minimo rischio con l'algoritmo Dijkstra applicato alla mappa di rischio generata Funzionamento: "algoritmo statico che calcola il percorso di minimo rischio su una mappa in cui

per ogni cella c'è un livello di rischio associato." —> risultato: mappe sulla provincia di torino ed oltre con percorso a minimo rischio. * ref. 8 il lavoro di Romeo e Cestino * INDICE DI MERITO: * 3 condizioni: payload, lunghezza del tragitto, sicurezza (nel caso del drone molto più penalizzante) * calcolo dell'indice di merito cambia dipendentemente dalla tipologia di missione (urgente, media, routine), pg. 86

* PRECEDENTE AD ANGELINI - TESI FIORENTINO 2021 * path-planning based on RRT* Method * concept of "layers" as in factors that need to be combined to have a full picture of the risk analysis * primatesta cited again (pg 40 explanation)

3.2. Proposed approach

3.2.1. (Outer) design loop formalisation

Aeronautical-projects inspired design loop WITH shift on payload-centric design, "protect the cargo rather than the machine" (rather new for the aerospace sector, where cargo and plane are NOT handled separately like in this case). Idea of the airbag.

3.2.2. Proposed approach

Inner design loop formalisation: specific requirements and design steps The diagram + objectives, i.e. where this MSc Thesis "stops" (slide 8 of Mid-term meeting pptx)

4

Results - Through the Workflow

Do not include too many implementation details

4.1. Mission and failure scenario analysis: calculating V_x and V_y

* Mi servono degli input: prendo le informazioni raccolte nella fase preliminare di analisi missione (ref analisi missione) per stabilire probabilità di impatto and what not.

* traiettoria * punto di partenza, punto di arrivo * drone scelto * payload scelto * medicinali scelti per simulazione

* serve energia di impatto?

—> risultati del WP2

4.2. Studying the payload: calculating allowable impact acceleration

4.2.1. Material characterisation

TEST

FEM

MENTION MODEL ITERATIONS TRIAL-AND-ERROR IN THE APPENDIX B!!!! FROM ABAQUS IMPLICIT TO CURRENT DECISIONS. Step by step, as in "version 1, abaqus implicit". Version 2, ls-dyna tetra elastic. Etc. MODAL PERTURBATION

* impact of container in LS-Dyna and what not. * diversi angoli di assetto per vedere il caso peggiore di accelerazioni locali * cosa possono sopportare le fiale e il contenitore???? * tipi di impatto, contenitore tipico che le contiene (menzionare la tesi del tipello) e che per il dimensionamento non viene considerato, approssimato a infinitamente rigido e che trasmette le accelerazioni di impatto così come le riceve —> airbag deve essere dimensionato così —> tutte le analisi statistiche di prima, che hanno tenuto in considerazione anche il fatto che prima o poi si sorvolerà la città una volta risolti i cavilli burocratici che impongono una certa mitigazione del rischio —> airbag.

—> risultati del WP3

—> CONCLUSIONE: I have what I need to design the damn airbag.

4.2.2. Sled test

4.3. Design variables optimisation: the code!!!

* codice ispirato da Zhou et al 2019 - spiegazione del modello e dell'aspetto dei vari blocchi * Sapendo che le thresholds delle provette sono tot, allora si può convergere verso un pre-dimensionamento dell'airbag

con tot dimensioni o caratteristiche. * oracle code for direzionamento verso un punto di partenza per il design

* I CHOSE THE CRUDE IMPLEMENTATION — THIS IS BECAUSE THE ONLY COUPLING ANALYTICAL MODEL IS MISSING AND I CANNOT RETRIEVE IT. THEREFORE THIS IS MY IMPLEMENTATION XXX Title: "Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers"

Author: Timothy F. Patterson

Report: NATICK/TR-87/042 (DTIC accession ADA199593)

Date: August 1985 (work performed March 1984–May 1985)

4.4. Summary of results

Actual design variables p inflation, diameter, shape, etc

****RESULTS: BRING IT BACK TO THE RESEARCH QUESTIONS!!!!****

5

Conclusions

5.1. General summary

Bring back the idea of the safety-first workflow into the broader context of the long-term, drone-network-all-over-Italy vision.

5.2. Recommendations for future work

Feeding these results in the “outer” design loop, updating the hardware and mission design, re-evaluating initial conditions to re-inform the “inner” design loop -> once reached convergence, the need to simulate and test the design implemented on the drone, with informed decisions on inflation mechanism and folding method.

combination of parachute (for, let's say, half the falling height) and then airbag for the second half? the objective is to minimise overall mass.

6

Project Management

Inspired by the Gibb's reflective cycle (?)

6.1. Strategy and planning

planning strategy (especially at the beginning), 5 supervisors to update independently, different means of updates - weekly meetings that became only when i had updates - regular bi-weekly meetings - irregular meetings and informal discussions here and there because of huge workload

6.2. Challenges

6.2.1. challenge 1: aligning interests for thesis scope

- 5 supervisors with 5 different visions of my thesis work. what contributed was a lack of a unified platform to check my progress and congregate materials.

6.2.2. challenge 2: Computational resources NO MORE

- by contract, politecnico should have given the resources, hpc and licences - unexpected maintenance, with delay - Bianca Ex Machina, access to DelftBlue. Out of contract, otherwise the thesis should have been changed

6.2.3. challenge 3: IT infrastructure bottleneck

- resulting project flow from company pc to personal pc, because no permission to install a VPN and launch from the company's computer.

6.2.4. challenge 4: going against the judgement of supervisors

- scegliere shell invece che solids in case of few supervisor - scegliere full invece che reduced model in case of tudelft supervisor.

6.3. Lessons learned

I'm really mad

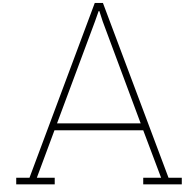
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Extended Market Analysis

A.1. Value-chain actors

Table A.1: Actors involved in drone delivery of medical goods in Italy, grouped by value-chain stage. [†] Based in Piedmont.

Category	Institution	Scope
Drone manufacturers	SkyEU [†]	Autonomous high-speed eVTOL drones for first aid and critical logistics [18]
	FlyingBasket	Cargo drones; used for blood transport Pozzuoli–Ischia [29]
	H2C	Hydrogen drones and hydrogen supply system for Padova Sandbox [30]
	Dronus	“Radon” drone for medical delivery [31]
	Eurolink Systems	“Beluga” medical delivery drone, with Università Cusano [31]
	IDS Ingegneria Dei Sistemi	Defibrillator-carrying drone (WingBeat project) [32]
Technology partners (role not specified)	Mavtech	INDOOR technical partner; aerial surveillance and operational-support products [19]
	FEV Italia [†]	Industrial partner of the DIMEAS feasibility studies; numerical impact analysis
	Leonardo	VTOL drone demonstration flights in Lazio, with Telespazio and D-Flight [33]
	Caltec	Technical partner of SIS 118 for the SEUAM defibrillator drone [34]
	AP 74 UAV	Drone company involved in the Piacenza pharmacy link [35]
Payload / container makers	ABzero	Smart Capsule: sensorised, UN3373-certified container for blood, organs and drugs [20]
	Carpitech	“Philotea” temperature- and atmosphere-controlled capsule [36]
	Air Abruzzo	“Flying Box” transport capsule [36]
	Telespazio	TPZ Air 100 drug box used in U-ELCOMÉ [37]

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Category	Institution	Scope
Operators	ABzero	Drone operator for San Raffaele trials; handled ENAC authorisation for INDOOR [23], [38]
	Unmanned4you	Operated Aetos blood-transport flights with FlyingBasket [29]
	DL Droni	Piloted the Piadena pharmacy flights [35]
	Telespazio	Runs the U-ELCOME hospital-to-village drug delivery service [16]
	Croce Rossa Italiana	Own drone fleet and certified pilot school; urgent-drug delivery trials [39], [40]
Airspace, infrastructure and energy	ENAV	Air navigation service provider; coordinated Corus-XUAM [41]
	D-Flight	U-space services (ENAV group); Corus-XUAM partner [41]
	DTA	Corus-XUAM partner, Grottaglie airport test site [41]
	ADPM Drones	“Hangar”: docking and recharging station for automated missions [41]
	Gruppo SAVE	Airport group leading Padova Sandbox [30]
	Heron Air	Operator of Padova airport, Sandbox partner [30]
	Milani; DBA Group	Build the hydrogen production and refuelling unit for Padova [42]
	Stesi	Supervision system, ground logistics and spare-parts analytics for Sandbox [43]
	Torino Caselle and Cuneo Levaldigi airports [†]	Proposed hub and support base for hydrogen drone logistics [24]
Research	PIC4SeR, Politecnico di Torino [†]	Autonomous flight technologies for transporting biological material (INDOOR) [25]
	DIMEAS, Politecnico di Torino (Prof. E. Cestino) [†]	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost-safety comparison of car, drone and eVTOL, hub siting [9], [10], [11], [12], [13], [14]
	Università di Torino [†]	INDOOR project partner [19]
	Scuola Superiore Sant’Anna	Parent institution of the ABzero spin-off [23]
	Università Cusano	Co-developer of the “Beluga” drone [31]
	San Raffaele technology centre	Developed the FF2020 drone delivery demonstrations [44]
End users	AOU Città della Salute e della Scienza [†]	INDOOR test flight between hospital sites; proposed hydrogen logistics plan; pharmacy department partner of DIMEAS feasibility studies [21], [24]
	Regional Transplant Centre Piemonte–Valle d’Aosta [†]	INDOOR partner for organ and biological transport [19]
	Ospedale San Raffaele	In-campus delivery of drugs and biological samples [45]
	ASST Sette Laghi	Feasibility study for inter-hospital drug transport [46]

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Category	Institution	Scope
	Piadena community pharmacies	Pharmacy-to-pharmacy drone link [35]
	ASL 1 Abruzzo	Hospital-only drugs delivered to remote villages [16]
	ASL Napoli 2	Blood transport to Ischia [29]
	Ospedale Miulli	Demonstration flight of life-saving drugs [47]
	ASL Lecce	Planned Gallipoli–Casarano hospital link [48]
	Ospedale di Patti	Planned link to the Aeolian Islands [48]
	Ospedale Pediatrico Bambino Gesù	Partner in VTOL demonstration flights in Lazio [33]
	SIS 118	Emergency service promoting the SEUAM defibrillator drone [34]
Promoters and funders	Fondazione DOT [†]	Promoter of the INDOOR project [19]
	Centro Nazionale Trapianti	Funds the INDOOR research grant at PIC4SeR [22]
	City of Turin [†]	INDOOR partner; funded the LastMed call [19], [26]
	Regione Veneto	Signatory of the Padova Sandbox agreement [30]
	Regione Lazio	Partner in VTOL demonstration flights [33]
	Regione Campania	Institutional support for the Aetos trial [49]
	EU research programmes	H2020 Flying Forward 2020; U-ELCOM, co-funded by Horizon [38], [50]
	Abruzzo municipalities and Parco Sirente Velino	Host communities of the U-ELCOM service [16], [37]
Regulators and authorities	ENAC	National civil aviation authority; flight authorisations; protocol with the Red Cross [51]
	EASA	EU rules for drone operations, Regulation (EU) 2019/947 [52]
	Regione Piemonte [†]	Regional health authority; drone trials subject to benefit–risk analysis [27]
Alternatives and exits	Tecnocad / Tecnomill / Bylogix [†]	LastMed autonomous ground rover for last-mile drug delivery [28]
	Amazon Prime Air	Halted drone delivery tests in Italy [53]

A.2. Trials and planned initiatives

Table A.2: Drone delivery of medical goods in Italy: trials and planned initiatives (status as reported by the cited sources; checked 5 October 2026).

Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
Piedmont	Turin, Molinette ↔ San Giovanni Bosco feasibility studies	2020–2024	DIMEAS (Politecnico di Torino); AOU Città della Salute e della Scienza (Pharmacy); FEV Italia	Life-saving drugs, blood-derived medication	Approx. 10 km corridor along the Po, approx. 15 min at 17 m/s	Desk studies (5 MSc theses, 1 conference paper); no flights performed	[9], [10], [11], [12], [13], [14]
Piedmont	Turin, INDOOR	Oct 2024	Fondazione DOT; PIC4SeR (Politecnico di Torino); ABzero; AOU Città della Salute e della Scienza	Biological samples (blood) in sensorised capsule	CTO → Molinette, 500 m, automatic navigation over the city and the Po	Single experimental flight	[19], [21]
Piedmont	Turin, hydrogen drone logistics plan	Apr 2025	AOU Città della Salute e della Scienza (Commissioner T. Schael)	Drugs, small medical loads	Torino Caselle hub, Cuneo Levaldigi support, future Parco della Salute	Proposal; no implementation found	[24], [54]
Abruzzo	Avezzano → Collelongo, U-ELCOME	Jul 2025	Telespazio (Leonardo/Thales); ASL 1 Abruzzo; municipalities of Collelongo, Montereale, Rocca di Mezzo	Hospital-only drugs for frail patients	Hospital to village pharmacy, approx. 20 min	Start of operational phase; further routes announced	[16], [17]
Apulia	Grottaglie → Manduria, Corus-XUAM	by 2022	ENAV; D-Flight; DTA	Medical material	Airport to hospital	Test (reported in 2022)	[41]
Apulia	Taranto, SEUAM	Feb 2024	SIS 118; Caltec (Campanian aerospace consortium)	Automated external defibrillator	Urban, simulated cardiac arrest	First test of experimental project	[34]
Apulia	Lecce, ASL Lecce	2025	ASL Lecce	Not specified in source	Gallipoli–Casarano hospitals	Intended	[48]
Apulia	Acquaviva delle Fonti, Ospedale Miulli	Nov 2025	Ospedale Miulli	Life-saving drugs, biomedical material	25 kg hexacopter; 25 km in 25 min	Operational demonstration	[47]
Campania	Pozzuoli → Ischia, Aetos	Feb 2025	FlyingBasket; Unmanned4you; ASL Napoli 2	Blood, biological samples	17 km in 15 min; 4 flights; 5 kg carried	Second trial phase	[29], [55]

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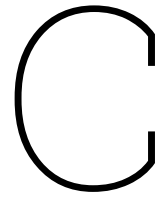
Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
Lazio	Rome	2022	Leonardo; Telespazio; D-Flight; Ospedale Pediatrico Bambino Gesù; Regione Lazio; Aeroporti di Roma	Not specified in source	Fixed-wing VTOL drones	Two demonstration flights	[33]
Lombardy	Piadena (Cremona)	2021	Two community pharmacies; DL Droni; AP 74 UAV	Drugs and devices between pharmacies	Approx. 6 kg electric drone, approx. 1 kg payload	Announced as continuous service; current status not verified	[35]
Lombardy	Milan, Ospedale San Raffaele, H2020 Flying Forward 2020	Mar 2023	Ospedale San Raffaele; EuroUSC Italia (regulatory consultant); ABzero (operator)	Drugs, biological samples	Within the hospital campus	Demonstration campaign, each flight ENAC-authorised	[38], [45]
Lombardy	Varese, ASST Sette Laghi	2025	ASST Sette Laghi	Drugs, urgent materials	Between two hospitals a few hundred metres apart	Preliminary feasibility study	[46]
Sicily	Trapani, Croce Rossa Italiana	2021	Croce Rossa Italiana	Urgent drugs	Port quay → vessel; 5 min 30 s vs 35 min by sea	Simulation exercise	[40]
Sicily	Patti → Aeolian Islands	2025	Ospedale di Patti	Not specified in source	Patti–Lipari	Under development	[48]
Veneto	Padova, Padova Sandbox	from Mar 2025	Gruppo SAVE; ENAC; Regione Veneto; H2C/H2G; Heron Air; Milani; DBA Group; Stesi	Medical goods, biological samples (hydrogen drones)	Up to 100 km, 4–5 kg payload (stated)	Ground tests of hydrogen system; first flight expected autumn 2026	[30], [56]

B

FEM Iterations for glass phial model

Adding source code to your report/thesis is supported with the package listings. An example can be found below. Files can be added using \lstinputlisting[language=<language>]{<filename>}.

```
1 """
2 ISA_Calculator: import the function, specify the height and it will return a
3 list in the following format: [Temperature, Density, Pressure, Speed_of_Sound].
4 Note that there is no check to see if the maximum altitude is reached.
5 """
6
7 import math
8 g0 = 9.80665
9 R = 287.0
10 layer1 = [0, 288.15, 101325.0]
11 alt = [0, 11000, 20000, 32000, 47000, 51000, 71000, 86000]
12 a = [-.0065, 0, .0010, .0028, 0, -.0028, -.0020]
13
14 def atmosphere(h):
15     for i in range(0, len(alt)-1):
16         if h >= alt[i]:
17             layer0 = layer1[:]
18             layer1[0] = min(h, alt[i+1])
19             if a[i] != 0:
20                 layer1[1] = layer0[1] + a[i]*(layer1[0]-layer0[0])
21                 layer1[2] = layer0[2] * (layer1[1]/layer0[1])**(-g0/(a[i]*R))
22             else:
23                 layer1[2] = layer0[2]*math.exp((-g0/(R*layer1[1]))*(layer1[0]-layer0[0]))
24     return [layer1[1], layer1[2]/(R*layer1[1]), layer1[2], math.sqrt(1.4*R*layer1[1])]
```



Task Division Example

If a task division is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Table C.1: Distribution of the workload

Task		Student Name(s)
	Summary	
Chapter 1	Introduction	
Chapter 2		
Chapter 3		
Chapter *		
Chapter *	Conclusion	
Editors		
CAD and Figures		
Document Design and Layout		

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ciao